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Final report on principles of refrigeration and air conditioning

Application of high temperature heat pump

Team members and division of labor:

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Summary

High-temperature heat pumps not only convert electricity into heat, but use heat pump circulation to raise the low-temperature heat source or waste heat in the factory that is difficult to use to a temperature that can be used in the process. Compared with traditional boilers, the characteristic of high-temperature heat pumps is not combustion heat production, but "transportation heat" and "upgrade heat". Therefore, it has important application value in industrial energy conservation, waste heat recovery and process heat electrification.

In the process of sorting out the information, we found that high-temperature heat pumps have a wide range of applications, including food processing, papermaking, textile dyeing and finishing, chemical industry, district heating and data center waste heat reuse. If you just introduce it one by one according to the industry, it will easily become a scattered collection of information. Therefore, we finally chose to use "process temperature, waste heat source, operating time and economic conditions" as the main analysis framework to determine the situations in which high-temperature heat pumps are more suitable for introduction.

The literature discussion section mainly refers to four materials: IEA HPT Annex 58 Final Report explains the positioning of high-temperature heat pumps in industrial process heat carbon reduction; Jiang et al. compile the cycle architecture and industrial application cases of high-temperature heat pumps; Jouhara et al. discuss modeling methods, refrigerants and component reliability; IEA 2025 analysis points out that industrial heat demand below 200°C has the potential for electrified heating. After synthesizing this literature, we believe that high-temperature heat pumps are best suited to be first introduced into processes below 200°C, with stable heat demand and with a recoverable waste heat source. For ultra-high temperature processes, high-temperature heat pumps are currently more suitable as preheating, waste heat recovery or boiler auxiliary equipment, rather than completely replacing the main heat source.

1. Research motivation

If the industrial sector wants to reduce energy consumption and carbon emissions, the difficulty is often not just "using less electricity" or "switching to green electricity", but the process itself requires a large amount of stable heat. Food processing requires sterilization and cleaning, papermaking requires drying, textile dyeing and finishing requires hot water and setting, and chemical processes also require reaction tank heating, distillation preheating or low-pressure steam. Most of this heat energy was supplied by natural gas, oil or coal-fired boilers in the past. Therefore, as long as factories rely heavily on combustion heat supply, energy costs and carbon emissions will not be able to truly decline. The appeal of high-temperature heat pumps is that they can reuse heat sources in factories that are otherwise discharged or difficult to utilize. For example, cooling water, hot waste water, exhaust gas, condensation heat, etc., if only relying on ordinary heat exchangers, often cannot be directly returned to the process due to insufficient temperature. However, through the heat pump compression cycle, these low-grade heat energy can be raised to a higher temperature, and then hot water, dry air or part of the low-pressure steam can be supplied. In other words, high-temperature heat pumps are more like "thermal energy upgraders" rather than devices that simply burn heat to produce heat in the traditional sense.

When discussing high-temperature heat pump applications, we believe that we cannot just ask "which industries can use it", because some application possibilities can be found in almost every industry that requires heat. What is more important is to determine "what on-site conditions are suitable for introduction." After sorting out the literature and comparing cases, we converged the discussion direction into three main issues:

First, whether the heat required for the process mainly falls below 200°C, especially hot water, dry hot air or low-pressure steam in the range of 80°C to 160°C. Second, is there a stable and recoverable source of waste heat on site, such as cooling water, hot waste water, exhaust gas, or condensation heat. Third, whether the process heat demand is stable enough to allow the equipment to operate for a high number of hours instead of being idle for a long time.

Through such a framework, we hope that the report will not only introduce the application list of high-temperature heat pumps, but also clearly explain the conditions, limitations and future development directions for its suitable introduction.

2. Literature discussion

2.1 IEA HPT Annex 58: Integration of high-temperature heat pumps into industrial process heat

Annex 58 Final Report of IEA Heat Pumping Technologies TCP focuses on industrial process heat and sorts out the technical status, application potential and introduction conditions of high-temperature heat pumps. The report classifies high-temperature heat pumps into technology types that have been commercialized, are close to marketization, and still need to be developed and demonstrated, and discuss their application possibilities in different industries, heating temperatures, and process integration methods [1].

The report points out that industrial process heat has long relied on fossil fuel supply. Therefore, if electricity can be used to drive high-temperature heat pumps and recover low-temperature or medium-temperature waste heat in the factory, fuel use and process carbon emissions can be reduced. However, the introduction of high-temperature heat pumps cannot be judged only by the COP of a single machine or the maximum heating temperature. It is still necessary to consider the heat source temperature, process requirements, equipment maturity, on-site integration and economy.

This document is intended to establish the technical positioning of high-temperature heat pumps in industrial applications. It supports us to view high-temperature heat pumps as "waste heat upgrade and boiler auxiliary" technology, rather than equipment that directly replaces all boilers. The subsequent analysis of process temperature, waste heat sources, operating time and economic conditions also continues the discussion direction of industrial process integration in this article.

2.2 Jiang et al. (2022): Circulation architecture and temperature rise requirements

The review article published by Jiang, Hu, Wang and others in Renewable and Sustainable Energy Reviews summarized the cycle architecture and application cases of industrial high-temperature heat pumps. This article discusses system forms such as single-stage compression, multi-stage compression, cascade cycle, mixed cycle, and mechanical vapor recompression, and compares the applicability of different architectures under heat source temperature, heating temperature, and temperature rise requirements [2].

The study pointed out that the feasibility of high-temperature heat pumps should not only look at the maximum temperature at the heating end, but also consider the temperature rise between the heat source end and the heating end. If the heat source temperature is high and the target heating temperature is not too high, a single-stage compression cycle may be able to meet the demand; if the temperature rise is too large, multi-stage compression or cascade cycles may be needed to reduce the burden on the compressor and improve efficiency.

We quote this document to support the "temperature" judgment in subsequent analysis. The temperature here includes not only the temperature required by the process, but also the temperature of the waste heat source itself and the temperature difference between the two. Therefore, when evaluating fields such as food, papermaking, textile dyeing and finishing, or chemical industries, you cannot just look at how much heat is required for the process. You must also consider whether the on-site waste heat is enough to serve as an effective heat source.

2.3 Jouhara et al. (2024): Modeling, refrigerant and system reliability

A review article by Jouhara et al. in the journal Energy summarizes the basic principles, modeling methods, refrigerant selection and system component design of high-temperature heat pumps. The article points out that before the introduction of high-temperature heat pumps, models need to be used to estimate COP, annual energy savings and performance under different operating conditions, while taking into account the thermal stability of the refrigerant, compressor efficiency, heat exchanger pressure drop, lubricating oil properties and control strategies [3].

Under high-temperature operating conditions, the refrigerant, compressor, lubricating oil, seals and heat exchangers will bear higher loads. If only the ideal cycle or nominal COP is used for evaluation, the actual energy saving benefits may be overestimated. Especially in industrial sites, changes in heat source temperature, unstable loads, exhaust pollution or heat exchanger scaling may cause system efficiency to decrease.

From this literature we set and supplement the engineering limitations of high-temperature heat pumps in practical applications. When the technical limitations are discussed in the "Main Limitations" later, the views of this document will be extended to include refrigerant stability, compressor burden, heat exchanger maintenance and system control into the assessment, instead of only judging whether high-temperature heat pumps are suitable for introduction based on theoretical COP.

2.4 IEA (2025): Process heat and electrified heating below 200°C

The article Can low-temperature heat in factories be electrified competitively? published by the IEA in 2025 discusses whether the low-temperature heat demand in industry can be supplied through electrification technology. The article points out that in industries such as food, textile and machinery manufacturing, a considerable proportion of heat demand is below 200°C. This type of process heat is more likely to be supplied through low-carbon technologies such as heat pumps, electric boilers, solar thermal energy or thermal energy storage [4].

The article also compares electrified heating methods such as heat pumps and electric boilers. Because heat pumps transport thermal energy rather than directly converting electrical energy into thermal energy, they usually have higher efficiency; however, equipment and installation costs are high, and the economics are still affected by electricity prices, fuel prices, equipment utilization, and carbon emission costs. Therefore, whether high-temperature heat pumps are worth introducing, in addition to technical feasibility, operating hours and investment recovery conditions also need to be considered.

By citing this document, we further focus on the analysis direction of process heat below 200°C. It also shows that high-temperature heat pumps are more suitable for medium and low-temperature heat needs such as hot water, cleaning, drying, sterilization and low-pressure steam, rather than directly replacing the main heat sources of ultra-high temperature processes such as steel, cement or glass.

2.5 Literature synthesis: import judgment structure

Based on the above four documents, the application of high-temperature heat pumps needs to consider technical feasibility, process conditions and economic benefits at the same time. IEA HPT Annex 58 provides perspectives on industrial process integration and carbon reduction; the research by Jiang et al. illustrates the impact of cycle architecture and temperature rise on system efficiency; the article by Jouhara et al. adds engineering constraints such as refrigerants, compressors, and system reliability; the IEA 2025 analysis points out that process heat below 200°C is an application range with greater electrification potential.

Therefore, subsequent analysis organized the introduction conditions of high-temperature heat pumps into four items:

First, whether the heat demand of the process is mainly lower than about 200°C; second, whether there is a stable and sufficient temperature waste heat source on site; third, whether the temperature rise between the heat source end and the heating end is reasonable; fourth, whether the system can achieve a reasonable payback period between equipment cost, electricity price, fuel price, operating hours and carbon emission cost.

If most of the above conditions are true, high-temperature heat pumps are more likely to become the main heating or boiler auxiliary equipment; if the process temperature is too high, the waste heat source is unstable, the temperature rise is too large, or the operating hours are insufficient, it is more suitable as a local preheating, cleaning hot water supply, dry air inlet preheating or demonstration system.

Table 1. Main literature focus and adoption methods

Literature	focus of discussion	Views available for this report	Corresponding chapter
IEA HPT Annex 58 Final Report [1]	Industrial process fever, technology maturity, and barriers to introduction	High-temperature heat pumps need to be integrated with plant heat flow, power and boiler systems	2.1、3.2、4
Jiang et al. [2]	High temperature heat pump cycle types and industrial application cases	The greater the temperature rise, the more multiple stages, cascades or special circulation structures are needed.	2.2、3.1、3.4
Jouhara et al. [3]	Modeling methods, refrigerants, compressors and system components	Need to estimate COP and reliability before importing, rather than just comparing nominal temperatures	2.3、3.4
IEA 2025 [4]	Industrial heat requirements below 200°C and electrification economics	High-temperature heat pumps are suitable for introducing medium and low-temperature process heat first, and can form mixed heating with boilers and thermal energy storage.	2.4、3.3、3.5、4

Reference, data collection and analysis

3.1 Basic principles and COP of high-temperature heat pumps

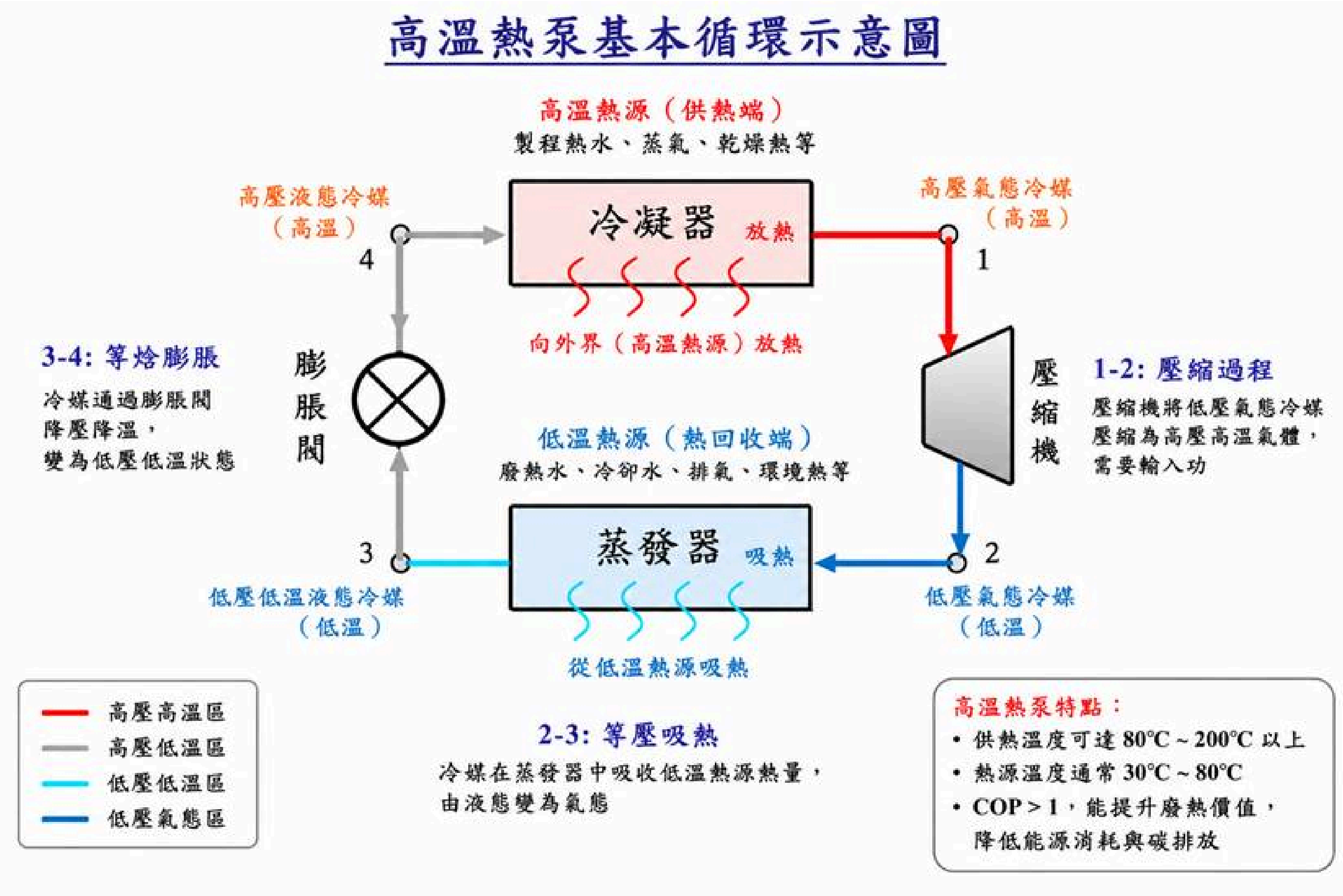

Figure ℧, basic cycle diagram of high temperature heat pump

High temperature heat pumps usually include an evaporator, compressor, condenser and expansion valve. The low-temperature heat source provides heat at the evaporator end to evaporate the refrigerant; the compressor inputs electrical power to increase the pressure and temperature of the refrigerant; the condenser releases heat to the process end; finally, the refrigerant returns to the evaporator after decompression through the expansion valve.

The performance of high-temperature heat pumps is often expressed as COP, which is the heat output divided by the work input. If COP = 3, it means that consuming 1 unit of electrical energy can supply about 3 units of heat energy. This is why high-temperature heat pumps are more attractive than resistance heating. Resistance heating roughly converts electricity directly into heat, while heat pumps use electrical power to transport heat, so when conditions are right, more heat can be provided with less electricity.

However, COP will be affected by the difference between the heat source temperature and the heating temperature. When the heat source temperature is high and the heating temperature is low, the pressure that the compressor needs to increase is relatively small, so the power consumption is low and the COP is high. On the contrary, if the temperature of the waste heat in the factory is low, but the process requires high temperature, the compressor must consume more electrical power, and the system efficiency will decrease. Therefore, when we evaluate a high-temperature heat pump, we should not only look at the maximum achievable temperature, but also the heat source temperature, heating temperature and annual operating hours.

3.2 Application mode

In practice, the introduction of high-temperature heat pumps usually does not completely replace boilers at the beginning, but starts with local processes or basic heat loads. After sorting it out, we believe that the import process can be roughly divided into seven steps:

Figure 2. High temperature heat pump introduction flow chart

It can be simplified to the following:

1. Look for sources of plant waste heat such as cooling water, exhaust, condensate water, hot waste water.
2. Confirm the temperature and heat required at the process end, such as hot water for cleaning, hot air for drying, and low-pressure steam.
3. Compare the difference between the heat source temperature and the heating temperature to determine whether single-stage compression, multi-stage compression or cascade circulation is suitable.
4. Evaluate electricity prices, fuel prices, equipment costs, maintenance costs and payback years.
5. Start with local processes, such as boiler feed water preheating, cleaning hot water or dry air inlet preheating, and then gradually expand.

A more conservative and common approach is to first use high-temperature heat pumps as boiler auxiliary equipment, such as preheating boiler feed water, providing hot water for cleaning, supplying part of low-pressure steam, or heating drying air. This strategy reduces the risk of introduction and allows the existing boiler to continue to provide backup during peak loads or high temperature demands. If the on-site waste heat is stable and the electricity price and carbon cost conditions are reasonable, the load ratio of the high-temperature heat pump can be further increased.

3.3 Industries suitable for import

Taking food and beverage factories as an example, there are often cooling and heating needs at the same time. For example, in the beverage manufacturing process, products need to be cooled or raw materials preserved, and hot water is needed for cleaning, sterilization, and equipment disinfection. If the condensation heat discharged from the cooling system is recovered and the temperature is raised through a high-temperature heat pump, part of the hot water demand can be supplied. The advantage of this application is that the heat source and heat demand are usually relatively stable, and the temperature mostly falls within the range that high-temperature heat pumps can easily achieve, so the introduction potential is high.

Textile dyeing and finishing plants often produce a large amount of hot wastewater. For example, the wastewater discharged after dyeing, cleaning and setting still contains considerable heat. If it is discharged directly, it will not only cause energy waste, but also increase the burden of back-end cooling and wastewater treatment. The high-temperature heat pump can recover the heat in the wastewater and reuse it for water replenishment heating in the dye vats or hot water for cleaning, reducing boiler steam usage. However, dyeing and finishing wastewater may contain dyes, fibers and chemicals, so the heat exchanger needs to consider scaling, corrosion and cleaning and maintenance issues.

The main characteristic of the papermaking industry is that the drying section consumes high energy, and the hot and humid exhaust gas and condensed water can be used as heat sources. If this heat can be recovered and used for dry air preheating or boiler feed water preheating, part of the fuel consumption can be reduced. However, the water vapor and dust in papermaking exhaust will increase the maintenance requirements of the heat exchanger, so the system design must consider filtration, cleaning and stability control.

Although the chemical industry also has medium and low temperature process heat requirements, such as reaction tank heating, distillation preheating and low-pressure steam, it has higher requirements for safety and continuous operation. Therefore, chemical industry sites are more suitable to start with preheating or boiler assistance, rather than letting high-temperature heat pumps replace the main heat source from the beginning.

In addition, data center waste heat is also a new application direction in recent years. The operation of the server will generate a large amount of stable waste heat. If it is heated through a high-temperature heat pump, it can supply building hot water, district heating or adjacent low-temperature processes. However, this type of application requires a heat demand end nearby and cannot affect the heat dissipation safety of the data center itself.

3.4 Main limitations

Although high-temperature heat pumps have the potential to save energy and reduce carbon emissions, not all sites are suitable for introduction. We organize the main limitations into three categories.

The first is technical limitations: high temperature and high pressure will increase the burden on the compressor, refrigerant, lubricating oil and seals. Reliability is very important during long-term operation. When the heating temperature increases, the compressor exhaust temperature may also increase. If the refrigerant or lubricating oil is not suitable, the life and safety of the equipment will be affected.

The second is economic constraints: the initial equipment cost of high-temperature heat pumps is relatively high, and whether it is cost-effective depends on electricity prices, fuel prices, operating hours and carbon fee systems. If the factory only uses heat energy occasionally, or if heat demand is unstable and equipment utilization is low, the payback period may be too long.

The third is the limitation of on-site integration: existing plants usually already have boilers, steam lines, heat exchangers and control systems. When introducing a high-temperature heat pump, the heat flow and pipelines need to be re-planned, which cannot be accomplished by simply buying a piece of equipment. If an energy flow inventory is not done first, equipment may be installed but not fully operational.

3.5 Economic benefit analysis

The economic benefits of high-temperature heat pumps mainly come from three parts: the first is to reduce the use of boiler fuel, the second is to recover the waste heat that was originally discharged, and the third is to reduce the cost of carbon emissions that may increase in the future. If the factory originally uses natural gas or oil-fired boilers for heating, high-temperature heat pumps can replace part of the fuel consumption with electricity. Although the price of electricity may be higher, because the heat pump does not directly convert electricity into heat, but transports heat energy, it may still have energy-saving benefits under conditions where the COP is greater than 2 or 3.

However, economic benefits cannot only be based on COP, but must also consider equipment investment, maintenance fees, annual operating hours and peak off-peak electricity prices. If the heat demand of the factory is stable and the operation hours are long throughout the year, the equipment utilization rate is high, and the investment payback period is usually short; if the heat demand is discontinuous and the heat pump is idle for a long time, the economic benefits will decrease.

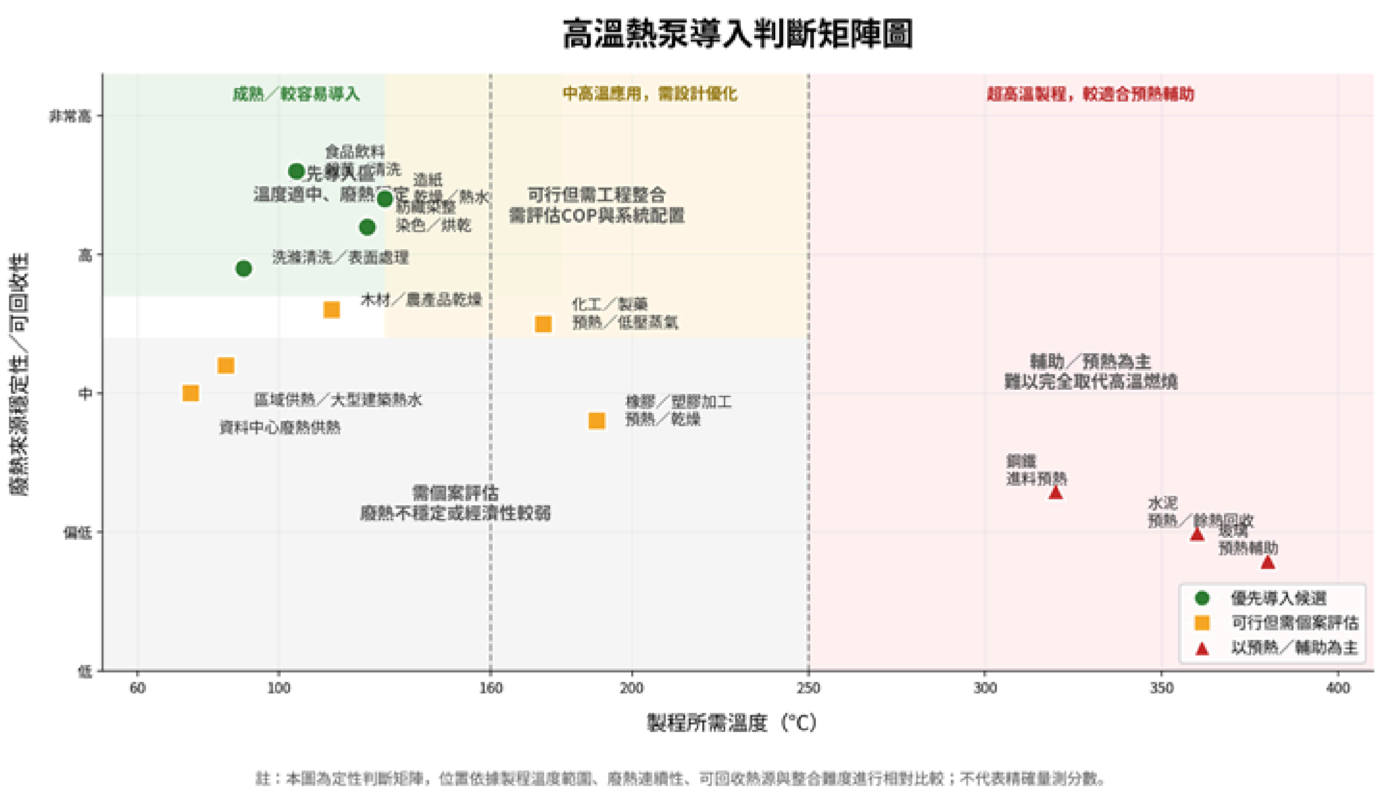

Figure 2. Judgment matrix diagram for high-temperature heat pump introduction (analyzed and compiled based on literature and this report)

It can be seen from the import judgment matrix that high-temperature heat pumps are most suitable for introduction in areas where the process temperature is not too high, the waste heat source is stable, and the recyclability is high. For example, industries such as food and beverage, textile dyeing and finishing, and papermaking are mostly located in more suitable areas; while ultra-high temperature processes such as steel, cement, and glass are more suitable for preheating or auxiliary systems even if there is a large amount of waste heat.

3.6 Practical application fields

In addition to using a matrix diagram to determine the feasibility of introduction in different industries, we also compiled the main heat requirements, recoverable heat sources and recommended uses of each application field into Table 2. This will provide a clearer comparison of why different industries are suitable or not suitable for the introduction of high-temperature heat pumps. The "import potential" in the table is not judged solely by the name of the industry, but rather comprehensively considers the process temperature, whether the waste heat source is stable, whether the heat demand is continuous, and whether the high-temperature heat pump can be integrated with existing boilers or process equipment.

Table 2. Comparison of high temperature heat pump application fields

field	Main heat demand	Recyclable heat source	Recommended use	Import potential
food and beverage	Sterilization, cleaning, cooking, drying	Cooling water, exhaust gas, condensation heat	Hot water, low pressure steam, dry hot air	high
paper making	Paper drying, process hot water	Hot and humid exhaust, condensate water	Dry air preheating, boiler auxiliary	high
Textile dyeing and finishing	Dyeing, cleaning, styling and drying	Hot wastewater, hot exhaust	Hot water supply, drying heat recovery	high
Chemical industry	Reaction tank heating, distillation preheating	Condensation heat, process waste heat	Process preheating, low pressure steam	medium to high
district heating	Building heating, domestic hot water	River water, sea water, sewage, data center waste heat	Central heating and hot water supply	medium to high
Steel/Cement/Glass	Ultra-high temperature process heat	High temperature exhaust gas	Feed preheating or auxiliary system	lower

To summarize briefly, food and beverage, papermaking, and textile dyeing and finishing are more suitable for the priority introduction of high-temperature heat pumps. The reason is that the heat needs of these industries are mostly concentrated in hot water, drying, and low-pressure steam, and there are often recoverable heat sources such as cooling water, hot wastewater, moist heat exhaust, or condensation heat. Chemical industry and district heating need to be evaluated based on process safety, heating network conditions and thermal stability. Although ultra-high temperature industries such as steel, cement and glass have a large amount of waste heat, the main process temperatures are too high, so high-temperature heat pumps are more suitable as preheating or auxiliary systems.

4. Conclusion and future prospects

The value of high-temperature heat pumps lies in upgrading low-grade heat energy, rather than simply adding a new heating device. From the literature and application fields, it is most suitable to be introduced into industries below 200°C, with stable heat demand and clear sources of waste heat. From the perspective of Taiwan's industrial environment, high-temperature heat pumps can be given priority in the introduction of hot water systems attached to food processing, textile dyeing and finishing, papermaking and electronics factories. These industries often have stable hot water, cleaning, drying or air conditioning and dehumidification needs, and may also have cooling water or heat exhaust sources.

In contrast, it is difficult for ultra-high temperature industries such as steel, cement and glass to directly replace the main heat source with high-temperature heat pumps, but they can still be used for feed preheating, low-temperature section heat recovery or auxiliary heat supply. This is also an important conclusion we got after sorting out the data: high-temperature heat pumps are not used to replace all boilers at once, but are more suitable to start with medium and low-temperature process heat and waste heat recovery to gradually reduce fuel usage. However, high-temperature heat pumps should not be overly idealized. If the process requires extremely high temperatures, or the temperature difference between the heat source and the heating end is too large, the system COP will decrease, and the equipment will also require higher-grade compressors, refrigerants, and heat exchangers. For existing factories, pipeline configuration, operation schedule, backup equipment and investment payback period often determine whether the introduction is successful. Therefore, the role that high-temperature heat pumps are really suitable for is as part of an industrial heating system, rather than replacing all existing equipment alone.

The future development direction can be roughly divided into four points. First, develop refrigerants with low GWP and suitable for high-temperature operation. Second, improve the reliability of compressors, lubricants and sealing materials under high temperatures and pressures. Third, develop multi-stage compression, cascade cycle and transcritical cycle to cope with the demand for large temperature rise. Fourth, combining thermal energy storage and smart control allows high-temperature heat pumps to adjust their operation according to electricity prices, heat demand and waste heat supply conditions.

Overall, high-temperature heat pumps are an important transition technology for industrial heating to shift from fossil fuels to electrification. Whether it can be successfully introduced not only depends on the equipment itself, but also depends on the process temperature, waste heat source, operating time, electricity price and on-site integration conditions. If introduced in a suitable site, high-temperature heat pumps can not only reduce boiler fuel consumption, but also further reduce process carbon emissions as electricity gradually decarbonizes in the future.

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